

Meridian Platform Runbook

On-call reference for the subscription and commerce platform.

Last reviewed 14 February 2025.

1. System Overview

The platform is a Python service backed by PostgreSQL 16 and Redis 7, deployed on three application nodes behind a load balancer. Static assets are served from a CDN. Background jobs run on a separate worker pool of two nodes.

The subscription biller is the most critical component. It runs nightly at 02:00 Pacific and charges every subscription due that day. A failed biller run is a Severity 1 incident regardless of time of day, because a missed run cannot simply be retried the next night without double-charging.

Service level objectives:

API availability	99.9 percent monthly
p99 request latency	400 milliseconds
Checkout success rate	99.5 percent
Biller completion	by 04:00 Pacific

The error budget is 43 minutes of downtime per month. When more than half the budget is consumed, feature deploys pause until the following month.

2. Environments

There are three environments: production, staging and development. Staging runs the same topology at one third the capacity and holds anonymised data refreshed weekly. Development uses synthetic fixtures only; production data may never be copied into development.

3. Deployment

Deploys go out through the CI pipeline on merge to main. A deploy takes roughly eleven minutes end to end and is a rolling restart, one node at a time, with a health check gate between nodes.

Deploys are frozen from 20:00 Pacific on Friday until 08:00 Monday, and during the last three days of any month, when subscription billing volume peaks.

To roll back, re-run the previous successful pipeline from the CI interface. Rollback takes about four minutes. Do not roll back by force-pushing to main.

Database migrations are applied separately from application deploys and always before them. Migrations must be backwards compatible for one release, so that a rollback of application code does not require a migration rollback.

Any migration expected to take longer than thirty seconds must be run outside business hours and announced in advance.

4. Monitoring

Dashboards are in Grafana. The four dashboards that matter on call are API Overview, Biller, Database, and Checkout Funnel.

Alerts route to PagerDuty. Severity 1 pages immediately, day or night. Severity 2 pages during business hours and queues overnight. Severity 3 opens a ticket and never pages.

Logs are retained for 30 days in the log store and for 400 days in cold storage. Traces are sampled at 5 percent of requests, raised to 100 percent automatically for any request that returns a 5xx status.

5. Incident Response

The first responder is whoever is paged. Their first duty is to acknowledge within five minutes, not to fix the problem.

For Severity 1, open an incident channel, post the current impact in plain language, and name an incident commander. The commander coordinates and does not debug. If the first responder is the only person available, they are the commander until relieved.

Declare Severity 1 when checkout is failing for more than 5 percent of attempts, when the biller has not completed by 04:00, or when the API is returning 5xx for more than 2 percent of requests for longer than five minutes.

Declare Severity 2 for degraded but working service: elevated latency below the 5xx threshold, a single failed worker node, or a delayed but progressing biller run.

Customer communication is the responsibility of the support lead, not the engineer. Do not post status page updates directly.

Every Severity 1 and Severity 2 incident gets a written postmortem within five working days. Postmortems are blameless and are published to the whole company. The action items are tracked in the normal backlog and reviewed monthly.

6. Database Operations

PostgreSQL runs as a primary with two streaming replicas, one in the same region and one cross-region. Replication lag is alerted above ten seconds.

Backups are taken as a full base backup nightly at 01:00 with continuous WAL archiving. The recovery point objective is 5 minutes and the recovery time objective is 15 minutes.

Restores are tested monthly against the staging environment. A restore that has not been exercised in the current quarter is treated as untested and may not be relied on in an incident.

Failover to the same-region replica is automatic after 30 seconds of primary unavailability. Cross-region failover is manual and requires director approval, because it cannot be reversed without data loss.

Never run a manual UPDATE or DELETE against production without a transaction and a colleague watching. Always run the SELECT form of the query first and confirm the row count.

Long-running queries above 60 seconds are terminated automatically. If a report needs longer, run it against the same-region replica instead of the primary.

Connection pool size is 40 per application node. Pool exhaustion presents as timeouts rather than errors, which is the most common cause of a confusing latency alert.

7. The Biller

The nightly biller is the component most likely to page you.

It runs at 02:00 Pacific, processes subscriptions in batches of 500, and writes an idempotency key for every charge attempt before calling the payment provider. That key is what makes a retry safe.

If the biller fails partway, do not restart it from the beginning without first confirming the idempotency keys are present for the completed batches. The restart command takes a batch offset for this reason.

A biller run that has not completed by 04:00 is a Severity 1. The usual causes, in order of observed frequency, are payment provider timeouts, connection pool exhaustion, and a migration that changed a column the biller reads.

Payment provider timeouts are retried three times with exponential backoff. After the third failure the subscription is marked for manual review rather than being retried indefinitely, and appears on the Biller dashboard.

Never issue refunds directly against the payment provider. Refunds must go through the admin interface so the ledger stays consistent.

8. Access and Security

Production access requires a hardware security key. Passwords alone are not sufficient and there is no break-glass password.

Access is granted by role, reviewed quarterly, and revoked automatically after 60 days without use. Requesting access back is a two-minute process, so the review errs on the side of revoking.

Secrets live in the secret manager and are injected at runtime. Secrets must never be committed to the repository, written to logs, or pasted into a chat channel. A secret that touches any of those is considered compromised and must be rotated the same day.

Customer payment details are never stored by Meridian. The payment provider returns a token and only the token is persisted.

9. Disaster Recovery

The disaster recovery plan assumes total loss of the primary region. Recovery runs from the cross-region replica plus object storage backups.

The full recovery drill is performed twice a year, in April and October, and is timed. The most recent drill, in October 2024, completed in 2 hours 40 minutes against a target of 4 hours.

The runbook itself is mirrored to a location outside the primary region, because a regional outage would otherwise take the recovery instructions with it.